SQL PORTFOLIO PROJECT

Customer Purchasing Behaviour

16 business questions answered across 3,863 Kenyan customer records

Mandela Chalingi Lotilo | SQL (MySQL) | KES

Executive summary

KES 29.92M
total revenue

KES 7,746
average purchase

79.9%
customers are Loyal (10+ purchases)

1.7%
gap: full price spends more than discounted

Growth lies in reaching more customers, not in raising spend per customer, which is almost uniform across every segment.

Business problem

Many MSMEs struggle to understand their customers' shopping behaviour well enough to improve sales, customer satisfaction and long-term loyalty from 2027 onwards.

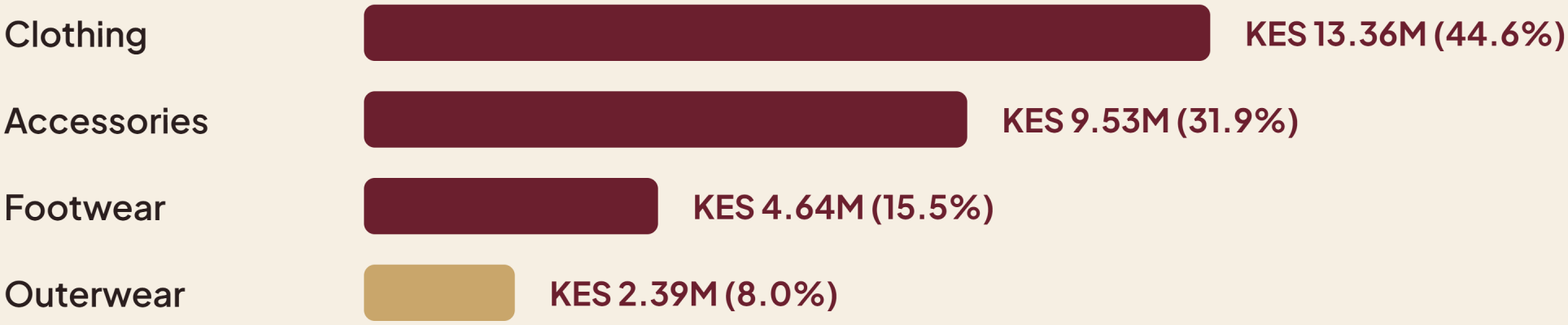
Management has noticed shifting purchasing patterns across demographics, product categories and sales channels (online vs. offline). They want to know which factors, such as discounts, reviews, seasons or payment preferences, drive consumer decisions and repeat purchases.

They tasked me with analysing the company's consumer behaviour dataset to answer one big question.

THE BIG BUSINESS QUESTION

How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

#1 Clothing and Accessories carry the business



Together, Clothing and Accessories deliver 76.5% of revenue.

#2 Revenue gap is a customer-count gap



about 2,615

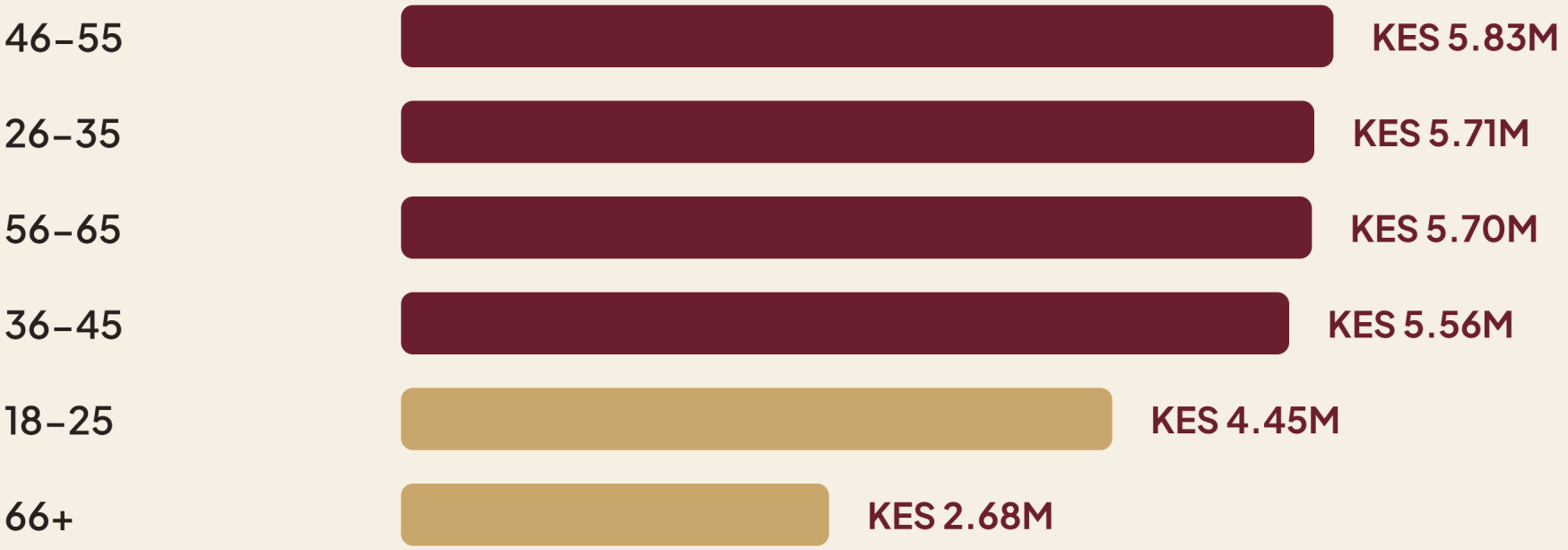
male customers, average spend KES 7,712

about 1,248

female customers, average spend KES 7,817

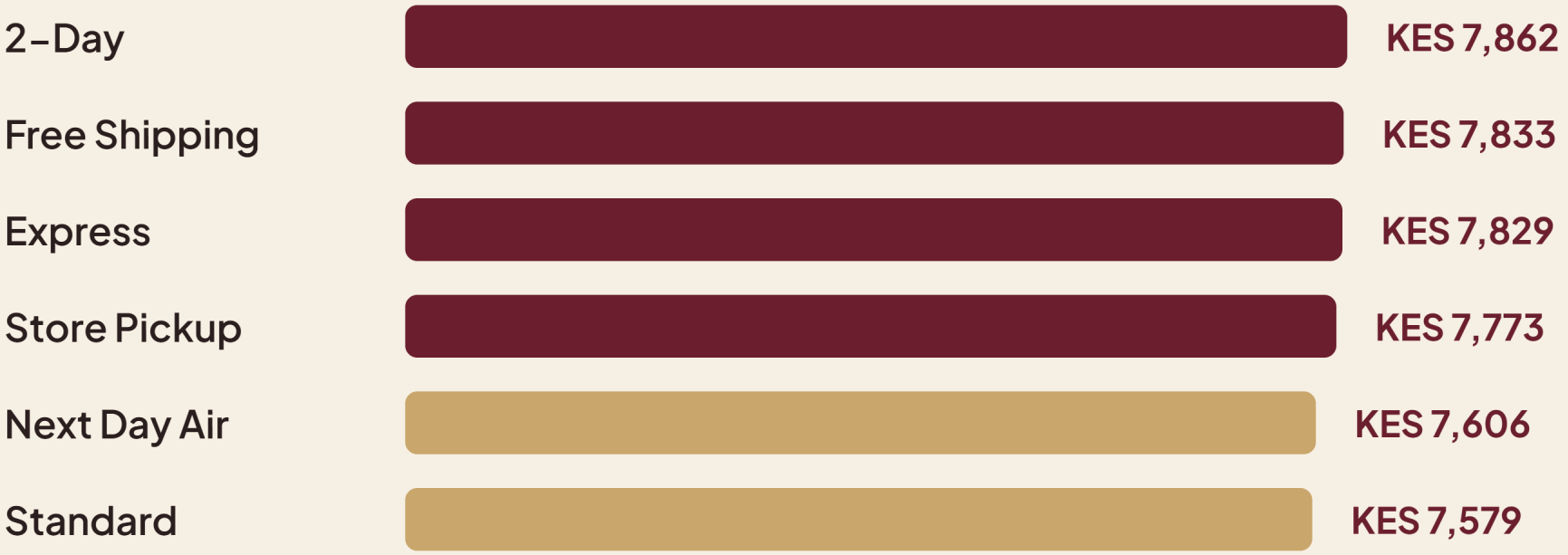
Women spend about 1.4% more per purchase, so the opportunity is acquiring more of them.

#3 Revenue by age group



Ages 26 to 65 each give 18.6% to 19.5%. Season is steady: Fall KES 7.74M leads Summer KES 7.15M by only 8.2%.

#4 Shipping choice barely changes spend



The spread from highest to lowest is KES 283 (3.7%), so shipping type is only a weak signal.

#5 Payment methods are spread thinly

673

PayPal customers, the most used method (17.4%)

112

Credit Card customers who buy every 3 months, the largest single group (2.9%)

Customers are spread across payment methods and purchase frequencies, so no single combination dominates.

#6 Discounts do not lift spending

KES 7,671

average spend with a discount (1,640 orders)

KES 7,802

average spend at full price (2,223 orders)

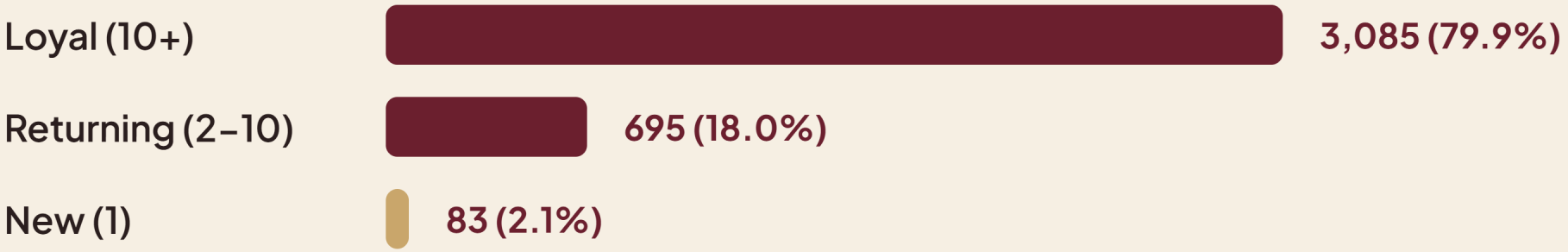
49.8%

of discounted customers (816) still spent above the KES 7,746 average

Top 5

discount-dependent: Hat 49.67%, Sneakers 49.66%, Coat 48.43%, Sweater 47.20%, Pants 46.43%

#7 A loyal base, few new customers



1,033 subscribed

Average KES 7,710, repeat rate 90.9%

2,830 not subscribed

Average KES 7,760, repeat rate 88.4%

Recommendations

Rethink discounts

Test targeting new or lower-spending customers; trial full price on Hat, Sneakers and Coat.

Grow female customers

They spend slightly more per purchase; reach more of them.

Back Clothing and Accessories

They deliver 76.5% of revenue; keep them well stocked.

Invest in acquisition

Only 2.1% of customers are new; judge subscriptions on retention.

Limits and next steps

- Observational data: associations, not proof of cause
- Small gaps untested for significance
- Amounts converted from the source dataset’s USD at 1 USD = KES 129.75 (1 November 2026)
- Next: Power BI dashboard, Python significance tests, customer lifetime value

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